W05 Requirement-to-Design Trace

ENGSE601 • Case X • TheOneBakery submission draft • Peer/source verification required

W05 draft for team review — this uses the official course HLD v0.9 Visual Guide plus the supplied S1 worksheet for the exact AC-X-05 / BR-X-02 descriptions. The linked full SRS/HLD DOCX files are listed as sources but were not parsed line-by-line in this preparation run; verify each trace against those full documents before final submission. No test results are claimed.

1. Scope and source register

Artifact	Version / location	How used / limitation
CaseX HLD v0.9 Visual Guide (official course portal)	https://se-rmutl.github.io/engse601/materials/casex/ENGSE601_CaseX_HLD_v0_9_Visual_Guide_TH.html	Official visual explanation of HLD Tables 1–5; component/layer/interface/decision descriptions. HLD is a draft under review, not necessarily correct.
CaseX SRS (official course portal)	https://se-rmutl.github.io/engse601/materials/casex/ENGSE_CaseX_SRS_v1_1_Practical_TH.docx	SRS IDs and acceptance basis. Exact SRS wording must be checked in the full DOCX; the course template README and portal mention v1.0/v1.1 respectively.
CaseX HLD v0.9 full draft	https://se-rmutl.github.io/engse601/materials/casex/ENGSE601_CaseX_HLD_v0_9_Draft_for_Review_TH.docx	Use for final source-level verification of tables and decision rationales.
Supplied S1 worksheet	ENGSE601_S1_Worksheet_TH.pdf, pages 1 and 4	Page 1 states AC-X-05 and BR-X-02 in the worksheet; page 4 is the bakery project's separate S1-B and is not used as Case X W05 evidence.

2. Architecture facts from HLD v0.9

The official visual guide describes four layers: Presentation (C-01/C-02), Application (C-03 to C-06), Integration (C-07 to C-09), and Data (C-10). C-01 displays a Readiness Summary and calculates Overall Readiness from items loaded for display. C-02 hides Decision buttons for items not owned by the signed-in owner and sends a decision to C-05. C-05 records a Decision Record, writes to C-10, and calls C-08 for notification. D-02 specifies a 15-minute cache; D-03 places Overall Readiness calculation in the UI; D-04 relies on hiding unavailable decision buttons.

3. Requirement-to-design trace

Trace / requirement basis	Design element and observable point	Status of design support	Fact / inference / missing evidence
TR-01 · FR-X-05 (linked by official visual guide to Overall Readiness; verify full SRS text)	C-01 Event Request UI (Presentation), table 2; D-03. Observable point: Readiness Summary shown in the request UI.	Partially Supported at HLD level	Fact: guide associates FR-X-05 with the C-01/D-03 Overall Readiness topic. Missing: exact SRS wording is not transcribed here; no explicit service-side enforcement point is described by D-03.
TR-02 · BR-X-01 (the guide labels this the overall-status rule)	C-01 calculates Overall Readiness from loaded items; C-03 handles request flow; D-03 leaves calculation at UI. Observable point: calculated status vs required-item approval state.	Partially Supported / risk open	Fact: guide maps BR-X-01 to Overall Readiness and cites a demo readiness service, but the HLD design decision D-03 says calculation is at the UI. Inference: multiple callers or stale loaded data could yield inconsistent status unless a single server-side rule is defined. Missing: approved central enforcement contract.
TR-03 · AC-X-05 (supplied worksheet, p.1): activity must not be Ready while a required item is unapproved	C-01 Readiness Summary / D-03. Observable point: attempt to derive or submit Ready when a required item remains unapproved.	Partially Supported; Not fully verifiable from HLD	Fact: exact acceptance statement comes from the supplied S1 worksheet; D-03 places aggregate calculation at UI. Missing: HLD does not specify server-side rejection, error contract, or a complete rule for every entry path. Test status: not executed in this W05 review.
TR-04 · BR-X-02 (supplied worksheet, p.1): a resource owner may decide only items assigned to them	C-02 Owner Review UI hides the Decision button for items not owned by the current user; I-04 sends decision to C-05; D-04 documents button hiding. Observable point: a decision request for a non-owned item.	Partially Supported; authorization boundary is unclear	Fact: UI-level hiding is documented. Inference: UI hiding alone can be bypassed if the service does not enforce ownership. Missing: explicit C-05 authorization check in HLD. The worksheet's direct-call question identifies this as a review concern, not a runtime exploit result.

4. Review findings — Fact / Inference / Missing Evidence

Finding	Fact (with source location)	Inference / impact on testability	Missing evidence
HF-01 · D-03 readiness logic	HLD Visual Guide, decision D-03: Overall Readiness is calculated in C-01 from loaded items.	Rule may be inconsistent if the status is trusted without a central service-side decision. AC-X-05 is difficult to verify at every entry path if only the UI calculation is defined.	Full SRS §/AC wording; service contract; server-side rule/owner confirmation.
HF-02 · D-04 owner decision	HLD Visual Guide, C-02 description + D-04: the UI hides Decision buttons for items not owned by current user; C-02 sends results to C-05.	A user may call a service directly unless C-05 independently checks the owner mapping and permissions. This is a design risk, not a confirmed runtime defect.	Full HLD wording for C-05, authorization contract, error response and permission test evidence.
HF-03 · D-02 cache	HLD Visual Guide, decision D-02: availability cache is 15 minutes.	Cached availability can be stale during the cache interval, which may affect readiness decisions. Actual acceptability depends on the SRS freshness requirement.	Exact linked SRS requirement, acceptable freshness, invalidation behavior and performance target.
HF-04 · HLD section 7 gaps	Visual Guide says interface formats/error codes, performance target TBD-X-01, notification channel/retry policy TBD-X-07, and retention/deletion policy TBD-X-08 are not specified.	These omissions can leave API/integration tests without stable oracles and leave operational risks unresolved.	Full HLD/SRS review and decisions by the requirement/design owner.

5. Design risk register

ID / level	Risk and root cause in design	Impact on testability	Proposed question/action	Owner/status
DR-01 / High	D-03 places Overall Readiness calculation in the UI.	Cannot prove AC-X-05 on all entry paths from UI behavior alone.	Should authoritative readiness be computed/enforced in an Application service? Define rule and observation point.	Design/requirement owner not assigned; Open
DR-02 / High	D-04 documents hiding decision buttons as the permission control.	Direct service/API permission bypass cannot be ruled out from HLD.	Confirm C-05 verifies owner mapping for every decision and returns a defined rejection response.	Design/security owner not assigned; Open
DR-03 / Medium	D-02 uses a 15-minute availability cache without linked acceptance threshold in this trace.	Stale data may change readiness; boundary of allowable staleness is not testable until specified.	Confirm SRS freshness requirement, cache invalidation and testable timing boundary.	Requirement owner not assigned; Open

6. Open questions and trace handoff

ID	Question requiring confirmation	Source / why open	Who should answer	Status / evidence needed
DQ-01	Which SRS baseline must this W05 cite: v1.0 or v1.1?	The current template W05 README names an SRS v1.0 file; course portal/visual guide describes SRS v1.1, and the visual guide says the IDs are the same.	Instructor / course source owner	Open — confirm baseline, then record exact section/text.
DQ-02	Must Overall Readiness be enforced in the service layer as well as displayed by C-01?	AC-X-05 on supplied S1 worksheet and HLD D-03.	Requirement/design owner	Open — decision and interface/rule evidence needed.
DQ-03	What must C-05 check when an owner submits a decision for a resource item?	BR-X-02 / C-02 / I-04 / C-05 / D-04.	Requirement/design/security owner	Open — permission rule and defined rejection behavior needed.
DQ-04	What freshness threshold and invalidation rule are required for cached availability?	HLD D-02 and missing target noted in HLD section 7.	Requirement owner	Open — SRS trace and decision needed.

7. Handoff / decision

Current W05 review result: NOT A SIGN-OFF. The available HLD material supports only partial traceability for readiness and owner-decision rules. Do not mark the design "Supported" merely because a UI component exists. Confirm exact wording and baseline in the full SRS/HLD DOCX, resolve owner questions, and update this PDF after peer review. This week is document review only: no Pass/Fail test result is claimed.

Evidence/terminology: TR = trace row; HF = finding; DR = design risk; DQ = open question. "Partially Supported" and "Not Verifiable" are design-review statuses, not runtime test results.

Sources used

1) Course HLD visual guide: https://se-rmutl.github.io/engse601/materials/casex/ENGSE601_CaseX_HLD_v0_9_Visual_Guide_TH.html

2) Course CaseX SRS v1.1 Practical: https://se-rmutl.github.io/engse601/materials/casex/ENGSE_CaseX_SRS_v1_1_Practical_TH.docx (referenced source; full DOCX wording must be checked).

3) Course CaseX HLD v0.9 draft: https://se-rmutl.github.io/engse601/materials/casex/ENGSE601_CaseX_HLD_v0_9_Draft_for_Review_TH.docx (referenced source; confirm table locations in full document).

4) Supplied ENGSE601 S1 worksheet, pages 1 and 4, used for AC-X-05 / BR-X-02 wording and to distinguish the separate THE ONE Bakery S1-B task.